Implement MINIMAX Algorithm

Aim :

To Use the MINIMAX algorithm to evaluate the optimal move in a game tree.

Procedure:

1. Define a list of terminal values (leaf nodes).
2. Apply recursive MINIMAX from root to leaves.
3. Alternate between max and min at each level.

Python Code:

```
def minimax(depth, nodeIndex, isMax, scores, h):
    if depth == h:
        return scores[nodeIndex]

    if isMax:
        return max(
            minimax(depth + 1, nodeIndex * 2, False, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, False, scores, h)
        )
    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, True, scores, h)
        )

scores = [3, 5, 6, 9, 1, 2, 0, -1]
h = 3
print("Optimal value:", minimax(0, 0, True, scores, h))
```

Output:

```
scores = [3, 5, 6, 9, 1, 2, 0, -1]
h = 3
print(&amp;quot;Optimal value:&amp;quot;, minimax(0, 0, True, scores, h))
Optimal value: 5
```

Result:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.